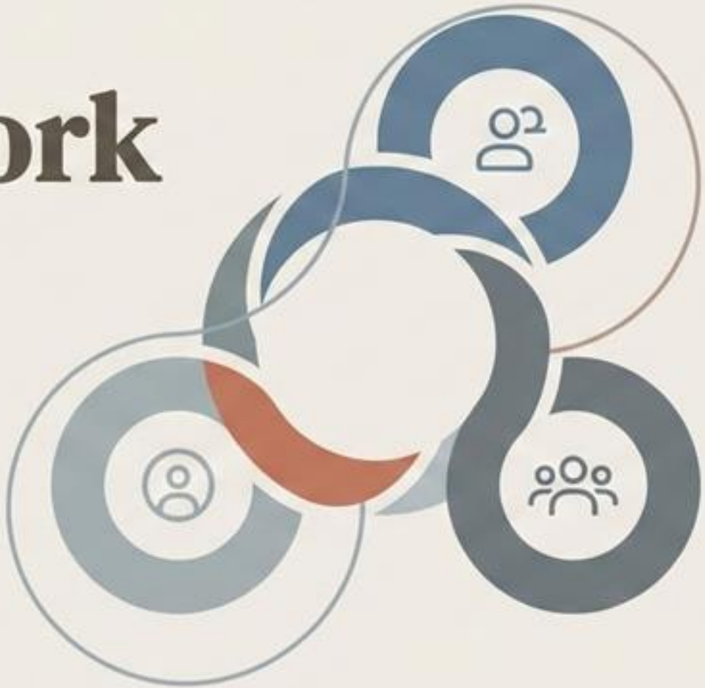
Scrum Framework and Roles

A Guide to Agile Project Management & Collaboration



Dr. Mohamed H. ElGazzar

Introduction to Agile and Scrum

Agile: The Mindset & Philosophy →

Agile is an iterative approach to project management and software development that helps teams deliver value to their customers faster and with fewer headaches.

- Customer collaboration over contract negotiation
- Responding to change over following a plan
- Working software over comprehensive documentation

Scrum: The Framework

Scrum is a lightweight framework that helps people, teams and organizations generate value through adaptive solutions for complex problems.

- Specific roles (Scrum Master, Product Owner, Team)
- Time-boxed events (Sprints, Daily Scrum)
- Defined artifacts (Product Backlog, Sprint Backlog)

What is Agile?

Agile is an iterative and incremental approach to project management and software development, prioritizing flexibility, collaboration, and customer feedback.

- Iterative Development: Breaking work into small, manageable chunks.
- Customer-Centric: Focus on delivering value to the customer.
- Adaptability: Embracing change and responding quickly.
- Collaboration: Emphasizing teamwork and communication.

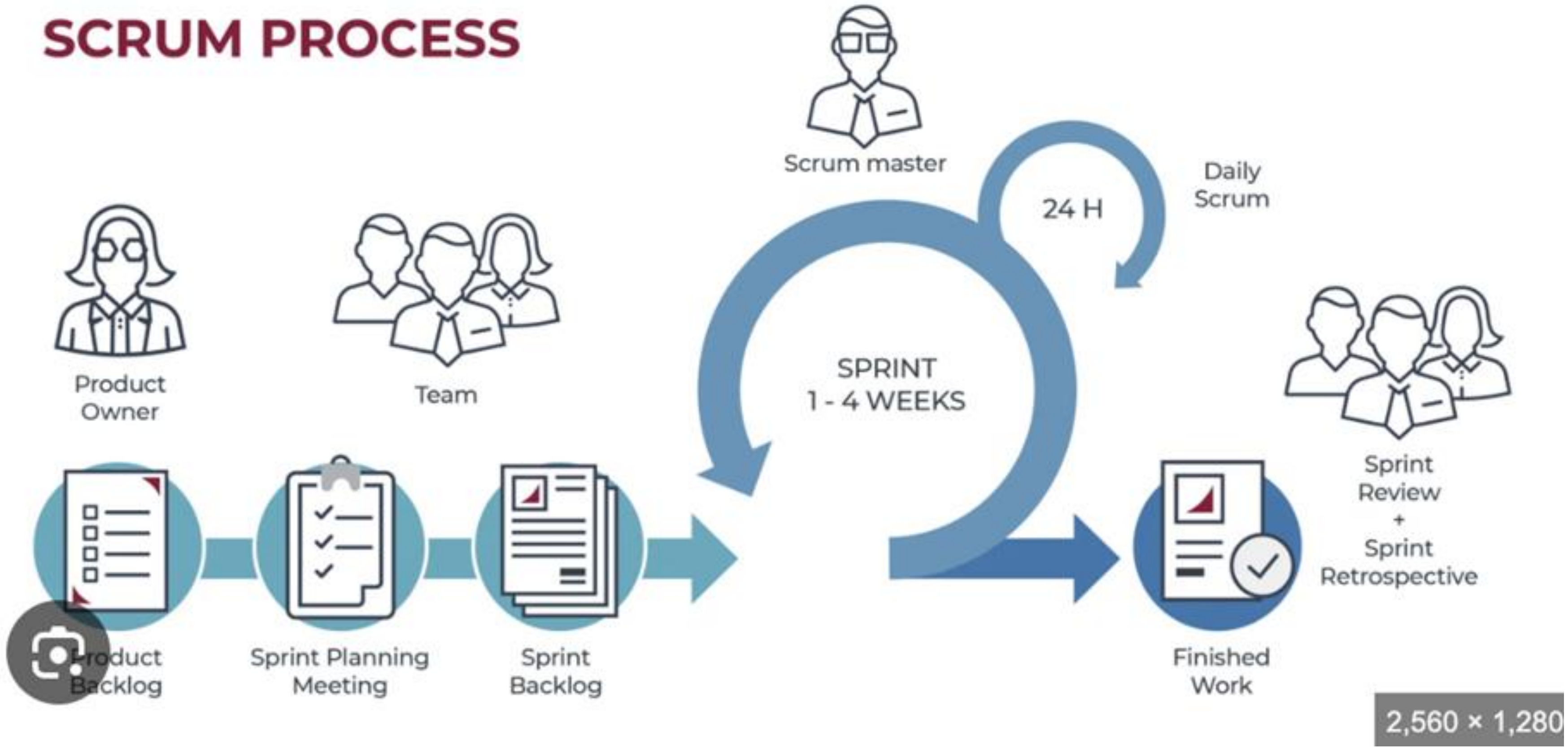
What is Scrum?

Scrum is a lightweight framework that helps people, teams and organizations generate value through adaptive solutions for complex problems. It is a widely used implementation of Agile.

- Framework, not a methodology: Provides structure for processes.
- Roles, Events, Artifacts: Clear responsibilities and time-boxed activities.
- Iterative & Incremental: Work is delivered in short cycles called Sprints.
- Transparency, Inspection, Adaptation: Pillars of empirical process control.

Scrum Cycle

- The Scrum Team completes the work in Sprints.
- Each Sprint results in a valuable Increment being produced.
- The Scrum framework uses a continuous cycle of events.



Source: https://ausum.cloud/scrum-metodologia-agil-mas-popular-en-empresas/

Three Pillars of Scrum

- Scrum is founded on transparency, inspection, and adaptation:

- **Transparency** ensures all aspects of the process are visible
- **Inspection** involves frequently checking Scrum artifacts and progress
- **Adaptation** means adjusting the process based on inspection results

Pillar 1: Transparency in Scrum

01 All aspects of the Scrum process must be visible.

02 The entire process must be observable by stakeholders.

03 Transparency requires common standards for work artifacts.

04 Decisions must be based on the perceived state of artifacts.

Pillar 2: Inspection

- Inspection frequency checks Scrum artifacts and progress
- Detect potentially undesirable variances and problems effectively
- Inspection process should not impede the actual work
- Frequent checking ensures alignment with goals and quality

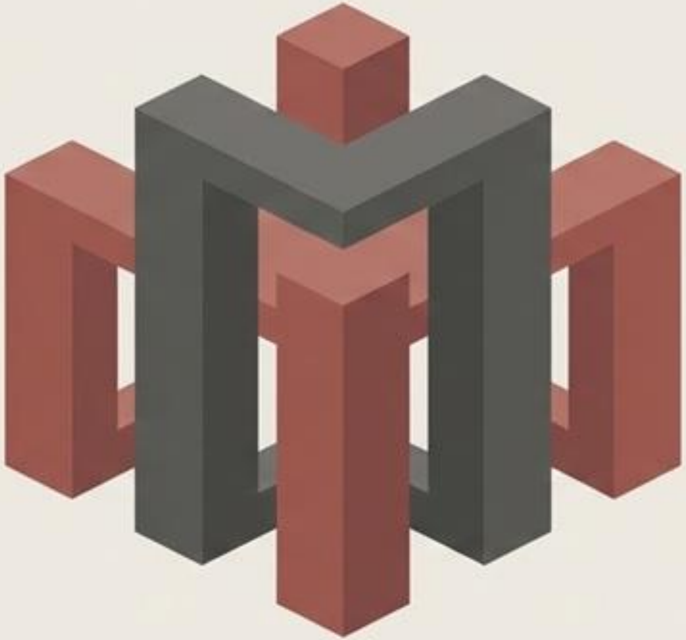
Pillar 3: Adaptation

01 Adjust the process or material when inspection reveals a problem

02 Adjustments must occur as soon as possible to minimize deviation

03 Adaptation primarily occurs during the formal Scrum Events

The Five Scrum Values



Commitment: People personally commit to achieving the goals of the Scrum Team.

Focus: Everyone focuses on the work of the Sprint and the goals of the Scrum Team.

Openness: The Scrum Team and its stakeholders agree to be open about all the work and the challenges.

Respect: Scrum Team members respect each other to be capable, independent people.

Courage: The Scrum Team members have courage to do the right thing and work on tough problems.

Scrum Value 1: Commitment

People personally commit to achieving the goals of the Scrum Team.

- Commitment to quality.
- Commitment to the Sprint Goal.
- Commitment to supporting each other.
- Commitment to continuous improvement.

Scrum Value 2: Focus

Everyone focuses on the work of the Sprint and the goals of the Scrum Team.

- Focus on the Sprint Goal.
- Focus on the highest priority items.
- Limiting work in progress (WIP).
- Minimizing distractions and interruptions.

Scrum Value 3: Openness

The Scrum Team and its stakeholders agree to be open about all the work and the challenges with performing the work.

- Openness about the work.
- Openness about the challenges.
- Open about progress and problems.
- Enables inspection and adaptation.

Scrum Value 4: Respect

Scrum Team members respect each other to be capable, independent people.

- Respect for people's skills and experience.
- Respect for different opinions and perspectives.
- Trusting the team to do the work.
- Respecting the timebox and agreements.

Scrum Value 5: Courage

The Scrum Team members have courage to do the right thing and work on tough problems.

- Courage to work on tough problems.
- Courage to speak the truth.
- Courage to be open and transparent.
- Courage to ask for help.

The Scrum Team

- The fundamental unit of Scrum is a small team of people.
- The Scrum Team *mainly* consists of the **Product Owner, Developers, and the Scrum Master**.
- There are no sub-teams or internal hierarchies within the Scrum Team.
- Scrum Teams are cross-functional and self-managing groups.

Cross-Functional and Self-Managing

- Teams possess all necessary skills to create value.
- Cross-functional teams have everything needed for a Sprint.
- Self-managing teams decide internally on execution methods.
- They determine who performs tasks, when, and how.

Scrum Team Size

- Scrum Teams are typically composed of ten or fewer people.
- Teams must be small enough to remain agile and effective.
- They must be large enough to complete significant work effectively.
- Teams that become too large should consider splitting into multiple teams.

Focused Team, Shared Goal

- The Scrum Team focuses solely on the Product Goal.
- They concentrate on only one objective at a time.
- Teams share accountability for producing a valuable Increment.
- A useful Increment must be created every single Sprint.

Scrum Roles

Deep Dive

Scrum Roles

- The Scrum Team includes the Product Owner, Developers, and Scrum Master
- The **Product Owner** is responsible for maximizing the product value
- **Developers** are the people committed to creating any aspect of a usable Increment
- The **Scrum Master** is accountable for establishing Scrum as defined in the guide

Product Owner (PO)

- The Product Owner is the visionary and customer voice.
- They maximize the value of the product from the Scrum Team's work.
- The method for maximizing product value varies by organization.
- The Product Owner ensures the most valuable work is done.

PO Accountability: Backlog

01 — The Product Owner must develop and clearly communicate the Product Goal.

02 — Ensure the Product Backlog (features) is transparent and understandable to everyone.

03 — Clearly communicate Product Backlog items to all stakeholders and the team.

04 — The Product Owner orders Product Backlog items by value and priority.

Product Owner Authority

- The Product Owner is a single individual, not a committee.
- They may represent group desires in the Product Backlog.
- All priority change requests must convince the Product Owner.
- Organization-wide respect for decisions is key to their success.

Fitness App: PO Vision and Roadmap

- Product Owner defines the vision for the fitness application
- PO maximizes value of the product from the Scrum Team's work
- PO manages the Product Backlog, prioritizing features and goals
- Ensure the Product Backlog is transparent and understandable to everyone
- The Product Owner ensures the most valuable work is being done

Fitness App P1: Core User Journey

- Focus on the core user journey and tracking
- Key functionality includes user onboarding and goal setting
- Implement manual workout logging in the current sprints
- Basic timer and rest intervals are also a core priority

- **PO Rationale (why):** *Establishes the core loop. Users need a reliable way to track progress before we introduce* complex *features.*

Scrum Master Role

- Scrum Master establishes Scrum as defined in the Guide.
- They help understand Scrum theory and practices.
- Scrum Master is accountable for team effectiveness.
- They ensure proper communication of Product Goal.

Scrum Master: True Leader

- The Scrum Master is a True Leader who serves (servant leader).
- They manage the Scrum process, not the people involved.
- They help the team and organization understand and implement Scrum framework.

Scrum Master Services to the team

- Coaching team members in self-management and cross-functionality.
- Helping the Scrum Team focus on creating high-value Increments.
- Causing the removal of impediments to the team’s progress.
- Ensuring proper understanding of Scrum theory and practices.
- Protect the scrum team from any distraction during the sprint.

Examples of SM services to the team

- **Removing Impediments:**

Example: If the Developers are blocked because they need an API key from an external partner to integrate a new workout device, the Scrum Master will step in to follow up with the partner or the internal legal team to cause the removal of that bureaucratic or technical hurdle.

- **Coaching in Self-Management and Cross-Functionality:**

Example: When the team is deciding how to implement "Manual Workout Logging," the Scrum Master will coach them to self-organize and decide on the technical solution (e.g., front-end vs. back-end responsibilities) rather than dictating the method.

Examples of SM services to the team

- **Helping Focus on High-Value Increments:**

Example: If the Developers start over-engineering a low-priority feature, such as complex social sharing icons, the Scrum Master will remind them of the current Sprint Goal (e.g., completing the "Core User Journey") to help the team focus on creating the most valuable functionality first.

- **Protecting the Team from Distractions:**

Example: During the Sprint, if a senior executive or external stakeholder tries to pull a Developer into an unplanned meeting to discuss a future feature, the Scrum Master will step in to deflect the request and shield the team, ensuring they remain focused on the "P1: NOW" features.

Examples of SM services to the team

- **Ensuring Proper Scrum Practices:**

Example: Coaching the Developers on how to use the Definition of Done (DoD) to maintain quality, ensuring that every piece of the Increment (like the "Basic Timer" functionality) meets the agreed-upon standards before being considered complete.

Scrum Master Services to Product Owner

- Helping find techniques for effective Product Goal definition
- Assisting with efficient Product Backlog management techniques
- Helping the team understand clear and concise Backlog items
- Facilitating stakeholder collaboration as requested or needed

Fitness App: Example of SM services to PO

- Ensuring the Product Goal is clearly articulated as "To deliver a personalized fitness experience that keeps users engaged, motivated, and consistent with their health goals".

- Coaching the Product Owner on how to write Product Backlog items (like "Implement manual workout logging") so that they are transparent and clearly understood by the Developers.

- Facilitating meetings or discussions with stakeholders (e.g., marketing, executive team) to ensure their input is considered and aligned with the Product Owner’s decisions on the Product Backlog and priorities.

Scrum Master Services to the organization

Leading, training, and coaching the organization in Scrum adoption.

Planning and advising on effective Scrum implementations.

Examples of SM services to the organization

- **Leading, Training, and Coaching in Scrum Adoption:**

Example: Training the Marketing, Sales, and Executive teams on the principles of Scrum, specifically demonstrating how their feature requests or deadlines must flow through the Product Backlog and Sprint planning process. This helps them understand why they cannot make direct, ad-hoc requests to the Development Team.

- **Planning and Advising on Effective Scrum Implementations:**

Example: Advising the organization on which metrics (e.g., team velocity, release frequency) are effective indicators of the fitness app's development health, rather than relying on traditional project management metrics.

The Developers' Role

- Developers are committed to creating a usable Increment.
- They are the people who build every aspect of the Increment.
- Required skills for Developers vary by the product domain.
- Developers are part of the fundamental unit, the Scrum Team.

Developers' Accountabilities

- Developers create the Sprint Backlog plan.
- They must adhere to the Definition of Done for quality.
- Developers adapt their plan daily toward the Sprint Goal.
- They are professionals and hold each other accountable.